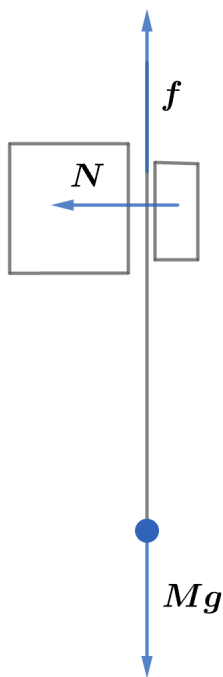


2011 F=ma Exam: Problem 16

Kevin S. Huang



To keep the rope from slipping, the friction force must balance the weight of Becky:

$$f = Mg$$

Since for static friction $f \leq \mu_s N$,

$$\begin{aligned} Mg &\leq \mu_s N \\ N &\geq \frac{Mg}{\mu_s} \end{aligned}$$

so this is the minimum normal force Jonathan needs to exert. Therefore, the answer is E.